When it comes to entertainment, you generally don't want to know what's happening off in the wings; you are perfectly content to focus solely on the production. But when it comes to understanding the universe, there is an insatiable urge to pull back all curtains, open all doors, and expose completely the deep inner workings of reality. Bohr considered this urge baseless and misguided. To him, reality *was* the performance. Like a Spalding Gray soliloquy, an experimenter's bare-bones measurements *are* the whole show. There isn't anything else. According to Bohr, there is no backstage. Trying to analyze how, and when, and why a quantum wavefunction relinquishes all but one possibility and produces a single definite number on a measuring device is missing the point. The measured number itself is all that's worthy of attention.

For decades, this perspective held sway. However, its calmative effect on the mind struggling with quantum theory notwithstanding, one can't help feeling that the fantastic predictive power of quantum mechanics means that it *is* tapping into a hidden reality that underlies the workings of the universe. One can't help wanting to go further and understand how quantum mechanics interfaces with common experience—how it bridges the gap between wavefunction and observation, and what hidden reality underlies the observations. Over the years, a number of researchers have taken up this challenge; here are some proposals they've developed.

One approach, with historical roots that go back to Heisenberg, is to abandon the view that wavefunctions are objective features of quantum reality and, instead, view them merely as an embodiment of what we know about reality. Before we perform a measurement, we don't know where the electron is and, this view proposes, our ignorance of its location is reflected by the electron's wavefunction describing it as possibly being at a variety of different positions. At the moment we measure its position, though, our knowledge of its whereabouts suddenly changes: we now know its position, in principle, with total precision. (By the uncertainty principle, if we know its location we will necessarily be completely ignorant of its velocity, but that's not an issue for the current discussion.) This sudden change in our knowledge, according to this perspective, is reflected in a sudden change in the electron's wavefunction: it suddenly collapses and takes on the spiked shape of Figure 4.7, indicating our definite knowledge of the electron's position.